Mill Plan

Milling sequence and setup instructions

Board Specifications

Dimensions: 12.000" (12") × 18.000" (18") × 1.500" (1 1/2")

Style: checkerboard

Species: Walnut, Maple

Milling Sequence

1. Mill rough stock

Joint one face and one edge on the jointer, then plane to final thickness plus 1/8" (3 mm). Rest overnight, then plane to final thickness. Take ~1/64" (0.4 mm) passes for the last two passes.

- Final pass depth: ~1/64" (0.4 mm)

2. Rip strips

Rip strips with the jointed edge against the fence. Measure from the fence face to the near side of a tooth set toward the fence.

- Fence to blade: strip width from cut list

3. Glue up

Glue the first edge-grain panel. Alternate clamps above and below the panel and check for flat with a straightedge.

- Clamp spacing: about 8" (200 mm)

4. Flatten first panel

Plane or sand the panel flat. It is still long grain, so the planer is allowed here if the panel is no wider than 12" (305 mm); a wider panel goes to the router sled or drum sander.

5. Crosscut slices

Crosscut the panel into slices of the final board thickness plus the end-grain flattening allowance (5 mm), on a crosscut sled with a stop block.

- Slice length: final board thickness from cut list

6. Rotate slices

Turn each slice end grain up.

7. Flip alternate slices

Flip every other slice end for end so the colours alternate into squares.

8. Glue up

Glue slices into the final panel, aligning the squares at every joint. Alternate clamps above and below the panel and check for flat with a straightedge.

- Clamp spacing: about 8" (200 mm)

9. Flatten and sand

Flatten with a router sled or drum sander, never the thickness planer. Sand 80, 120, 180 grit. Break edges (180 grit).

- Grits: 80 / 120 / 180

10. Edge profile

Rout the edge profile, default: 1/8" (3 mm) roundover. Optional finger grips: core box / round-nose bit 3/8-1/2" (9.5-12.7 mm).

- Bit: 1/8" (3 mm) roundover

11. Finish

Raise the grain with a damp cloth, sand 180 again, then flood with food-safe mineral oil and top with board butter.